

Multi-Alpha Quantitative Trading Lab: Technical Report

1. Executive Summary

This project constructs an end-to-end quantitative trading pipeline for single-instrument alpha discovery, strategy validation, and portfolio construction. The pipeline enforces strict research discipline: lookahead-bias prevention via structural timing enforcement, significance testing with autocorrelation adjustment and multiple-testing correction, and honest null-baseline validation for every learned model. The research finds no statistically validated edge in the supplied 986-day, single-instrument dataset using five distinct alpha strategies and five allocation methods, a null result that is itself the finding.

2. Data and Preprocessing

2.1 Raw Input

- **Price file:** 1000 OHLCV records, `price_train.csv`
- **Signal file:** 20 pre-computed indicators in three categories (price-based: PB01–PB08; band-based: BB01–BB07; volume-based: VB01–VB05), `signals_train.csv`
- **Period:** January 2, 2018 – December 2, 2021 (approximately 1000 trading days)

2.2 Data Quality Issues Discovered and Remedied

1. **Mixed date formats:** Most rows in `price_train.csv` use YYYY-MM-DD; ~30 scattered rows use DD-MM-YYYY (e.g., 21-02-2018). Parsed deterministically by detecting which token is 4 digits (`_parse_date_token` in `data_cleaner.py`), avoiding the ambiguity of `dayfirst` heuristics.
2. **Format-disguised duplicates:** 14 duplicate days in the price file, several hidden by using the *other* date format for the same OHLCV (e.g., 2018-07-18 and 18-07-2018 both present with identical prices). De-duplication runs *after* date normalisation so format-disguised copies are caught.
3. **Signal-only rows:** 14 dates present in the signal file with no matching price row (not duplicates, genuinely absent from the price file). Inner join on `date` drops these and logs the count; no forward-filling of prices that were never observed.
4. **Warm-up NaNs:** Columns BB06 (19 rows), BB07 (13 rows), VB05 (19 rows) have NaN at the start, representing rolling windows not yet filled. These are left as NaN (never fabricated) and strategies reading them are documented to treat NaN as “no signal.”

Final clean dataset: 986 usable rows (Jan 2, 2018 – Nov 1, 2021).

3. Backtesting Infrastructure (Task 1)

3.1 Core Architecture

The backtester enforces two critical invariants:

Invariant 1: No lookahead bias (t-1 cutoff) - Signals for decision at candle t may use data up to and including candle $t-1$'s close. - Implemented structurally: `Backtester.generate_signals()` shifts all signal columns by `SIGNAL_LAG=1` before passing to the strategy. The strategy never sees row t 's signals when deciding row t 's trade because they are not available. - Verified via two automatic tests: - *Truncation check*: `strategy.generate_signal()` run on `data.iloc[:k]` matches the full-sample run restricted to the same rows. - *Perturbation check*: Doubling a signal value strictly after the sample midpoint does not change any signal at or before the midpoint.

Invariant 2: Fill at open, cost charged correctly - All trades fill at candle t 's open price, not close. - Transaction cost: 0.05% per side (mandated by spec), applied to $|\text{position}(t) - \text{position}(t-1)|$. - Slippage: 5% of that candle's $(\text{high} - \text{low}) / \text{open}$ range, scaled by trade size (not mandated; documented assumption accounting for intraday volatility).

3.2 Accounting Convention: Open-to-Open

- Position held over $[t-1, t)$ interval: `position(t-1)` (entered at $t-1$'s open).
- Return earned: $\text{position}(t-1) \times (\text{open}(t) / \text{open}(t-1) - 1)$.
- Cost charged at t : transaction cost + slippage on the trade executed at t 's open.
- This matches the mandated fill-at-open rule exactly: the position change happens at t 's open, and its return (and cost) are realized immediately thereafter.

3.3 Baseline Strategy

BaselineStrategy: long when `PB01==1`, flat otherwise. Deliberately simple, exists only to verify the pipeline runs end to end. Over the 986-day sample: **Sharpe -0.23**, total return -13.2%, max drawdown -33.4%. Not profitable, as expected; this strategy reads a single discrete signal without any hypothesis beyond “does a simple trend flag predict anything.”

4. Alpha Research (Task 2)

4.1 Five Distinct Hypotheses

Each strategy encodes a different market mechanism and reads a mostly disjoint signal subset (enforced at the interface level):

4.1.1 Alpha01: Trend Agreement

- **Mechanism:** Trend continuation. Buy when short-term trend (PB01) *and* long-term trend (PB02) *and* price-above-average (PB04) all agree on direction.
- **Signals:** PB01, PB02, PB04 (all binary).
- **Reasoning:** Divergence between short- and long-term trends may be noise; agreement may indicate genuine momentum.
- **Result:** Sharpe -0.64 , permutation p-value 1.00 (observed Sharpe is at the left tail of the null).

4.1.2 Alpha02: Band-Extreme Mean Reversion

- **Mechanism:** Mean reversion at Bollinger Band extremes. Buy when overbought+confirmed, sell when oversold+confirmed.
- **Signals:** BB03 (overbought), BB04 (oversold), BB06 (continuous band position).
- **Reasoning:** Extremes tend to mean-revert before trending further; mean reversion typically has shorter-term validity than trend continuation.
- **Result:** Sharpe $+0.185$, permutation p-value 0.096 . Closest to significance (raw $p < 0.10$) but fails BH correction (adjusted $p = 0.575$ after testing 6 strategies).

4.1.3 Alpha03: Volume-Confirmed Breakout

- **Mechanism:** Long-only continuation on breakout + volume confirmation (no equivalent short rule found in signal library).
- **Signals:** PB06 (breakout), VB01 (volume participation), VB03 (volume-supported).
- **Reasoning:** Breakouts without volume are often false; volume-confirmed breakouts are more reliable.
- **Result:** Sharpe -0.58 , permutation p-value 0.70 .

4.1.4 Alpha04: Squeeze-Expansion Timing

- **Mechanism:** Volatility regime timing. Trade in momentum direction for 10 bars after a squeeze (low volatility, $BB05==0$ after $BB05==1$) ends.
- **Signals:** BB05 (squeeze flag), PB05 (momentum direction).
- **Reasoning:** Markets sometimes rally after a low-volatility regime ends; timing the regime switch and momentum direction may capture this.
- **Result:** Sharpe -1.39 , permutation p-value 0.96 .

4.1.5 Alpha05: Learned Composite

- **Mechanism:** Logistic regression on five continuous signals, predicting sign of 5-bar-forward price *difference*.
- **Signals:** PB07, PB08, BB06, BB07, VB05 (all continuous, ranges $\sim[-2, +2]$).

- **Model:** Logistic regression (6 parameters: intercept + 5 coefficients).
- **Model choice rationale:** ~600 development-period rows (before 2021-01-01). With 5 features, 6 parameters is at the ceiling of defensible complexity; a tree ensemble would carry hundreds of effective parameters against a few hundred rows and memorize noise. Predicting sign (direction), not magnitude (return), is consistent with the project discipline of using price *difference*, not level, as the prediction target.
- **Result:** Sharpe +0.102, permutation p-value 0.236. Balance report flags it: only 1 round trip over the sample, so win-rate (100%) and Sharpe are not statistically meaningful.

4.2 Signal Characterization

Hit-rate (direction-only) and information coefficient (magnitude-aware correlation) computed for all 20 signals against the forward price *difference* at horizons 1, 3, 5, 10, 20 bars. Top signals by absolute IC: - PB07 vs. 10-bar fwd diff: IC = -0.18 (significant correlation, opposite direction). - BB03 vs. 3-bar fwd diff: IC = -0.12 (mean reversion signal).

Individual signals show predictive power, but none of the five hand-built rules converts that into a cost-surviving edge after multiple-testing correction.

4.3 Statistical Significance: The Honest Null Result

Methodology Permutation test (corrected version): Circularly shifts the strategy’s *position series* against the fixed, real market data and re-runs it through the actual **ExecutionEngine** + **Portfolio**, computing the resulting Sharpe. Builds a null distribution of 500 permuted Sharpes. Tests whether the observed Sharpe is likely under the null; if not, the strategy’s edge comes from timing (being long/short at the right *times*), not from market drift or costs.

Newey-West t-stat: Adjusts the standard error of mean daily return for autocorrelation (positions persist, returns are autocorrelated); a plain i.i.d. t-test would be overconfident.

Multiple-testing correction: Benjamini-Hochberg (FDR control) across all 6 strategies tested. Per-test = 10%; after correction, the bar for any single strategy to be “significant” rises.

Results

Strategy	Sharpe	Hit-rate (5-bar)	Perm p-value	BH- adjusted p	Significant (=0.10)
baseline_pb01	0.23	48%	0.99	0.998	No
alpha_01	-0.64	42%	1.00	0.998	No
alpha_02	0.185	54%	0.096	0.575	No
alpha_03	-0.58	44%	0.70	0.998	No

Strategy	Sharpe	Hit-rate (5-bar)	Perm p-value	BH- adjusted p	Significant (=0.10)
alpha_04	-1.39	39%	0.96	0.998	No
alpha_05	0.102	50%	0.236	0.707	No

Finding: None of the six tested edges clears the corrected significance bar. Even `alpha_02`, which comes closest (raw $p = 0.096$, genuinely close to the conventional 0.05 line for a single test), is not significant after accounting for the six tests tried. This is reported as a result, not hidden.

4.4 Robustness Checks

Sub-period analysis (4 quarters, Sharpe per block): No strategy is stable across all four periods.

Cost sensitivity: For each strategy, re-run the backtest at cost levels 0%, 0.05%, 0.10%, 0.15%, 0.20%/side. None shows a plausible “break-even cost” much above 0.05%.

Regime conditioning (split by BB05 squeeze flag): Strategies do not concentrate their returns in one regime; results are not regime-specific artifacts.

Orthogonality analysis (pivoted QR on the return matrix): - Effective rank: 6 out of 6 strategies (full rank, no strategy is numerically redundant). - Rolling-window stability of the “top pivot” (most information-bearing strategy): 33% (changes often). The ranking of strategies is not a stable structural property on this sample.

4.5 Model Discipline: Every Model States Its Capacity Ceiling

`alpha_05`’s docstring and the project README state explicitly: “5 features, ~600 dev-period rows $\rightarrow$ 6 parameters (5 feature coefficients + intercept) is the defensible ceiling. More would memorize. This is a linear model because that’s how much structure the data can honestly support.”

Similarly, Task 3’s `AlphaMetaModel` (logistic regression on 2 features per strategy per rebalance window) includes: “~37 walk-forward folds, so 2 features $\times$ 3 strategies = 6 per-fold observations to build the feature matrix. A model with $> \sim 2$ parameters per strategy overfits by construction.”

5. Portfolio Construction & Allocation (Task 3)

5.1 Strategy Combination Problem

Task 3 needs at least two return streams to demonstrate allocation. Since Task 2 found no statistically validated edge, `practical_set_for_task3` carries forward the top-3 by unadjusted Sharpe: - `alpha_02_band_reversion` (0.185) - `alpha_05_learned_composite` (0.102) - `baseline_pb01` (-0.23)

Caveat: Explicitly stated in JSON output and console: “These three are un-validated. Allocation methods are demonstrated here; this is not a claim of alpha.”

5.2 Factor Model: Alpha vs. Beta Decomposition

Regression of each strategy’s return against a single factor: the asset’s own buy-and-hold open-to-open return.

Model: $r_i(t) = \alpha_i + \beta_i \times r_{BnH}(t) + \epsilon_i(t)$

Why this factor: Single-instrument dataset; no cross-sectional market index available. The asset’s own buy-and-hold return isolates how much of a strategy’s return is just “being long the underlying’s drift” (β_i) versus genuine timing skill (ϵ_i).

Results: - **baseline_pb01:** β = -2.07 (negative timing), $R^2 = 0.59$. Almost entirely beta; the return it achieved was just riding the market’s upward drift, with worse-than-passive timing on top. - **alpha_02_band_reversion:** β = +0.32 (insignificant), $\alpha = 0$, $R^2 = 0.0015$. Near-zero beta; return is not explained by buy-and-hold correlation. Mean reversion truly is a different bet. - **alpha_05_learned_composite:** β = insignificant, α = some positive value, $R^2 = 0.33$. Moderate beta; some drift correlation but also independent signal.

5.3 Five Allocation Methods Compared

Method	Mechanism	This Run Sharpe	Notes
1. best_individual	100% on single best Sharpe	0.185	Naive baseline; only alpha_02 gets capital.
2. equal_weight	1/3 each	0.009	Loses to the best single strategy due to including baseline_pb01.
3. inverse_volatility	Weight 1/ (ignores correlation)	0.031	Still loses; doesn’t account for co-movement.
4. mean_variance_shrinkage	Markowitz (Σ^{-1} shrinkage toward diagonal)	0.189	Best of five. Shrinkage prevents wild weights on a small sample (1000 bars, 3 strategies, ~singular covariance).

Method	Mechanism	This Run Sharpe	Notes
5. dynamic_meta_model	Walk-forward learned scores → softmax weights with 5% no-trade band	0.009	Converges to equal weight in practice due to the no-trade band + cold-start equal-weight fallback.

Key finding: Method 4 (mean-variance with shrinkage) edges out method 1 (best individual) — a modest gain from spreading risk more carefully — but method 5 (the learned allocator) shows *no validated benefit*. Its null-baseline check: - Real walk-forward direction accuracy: 52.3% (predicting sign of next month’s return for each strategy). - Permuted-label null: mean 48.2%, std 4.1% (100 permutations). - p-value: 0.198 (one-in-five chance a gap this size occurs in noise).

The learning did not produce a statistically significant improvement. The model’s architectural features (5% no-trade band, expanding-window walk-forward, cold-start equal weight) are sound, but the edge — if real — is too small to confirm on this data.

5.4 Rebalancing and Cost

- **Frequency:** 21 bars (~monthly), giving 37 walk-forward folds over the sample.
- **Static methods** (1–4): Fit once on the full return matrix, held fixed. No rebalancing cost.
- **Dynamic method** (5): Rebalances at each fold; charged 0.05%/side on each reallocation (same as any trade in the underlying).
- **No-trade band:** 5% absolute; weights only move if their change exceeds this band. Prevents churn on small, possibly-noisy model output changes.

6. Disciplines and Invariants

The entire project enforces five core disciplines:

1. **Price difference for prediction, not price level.** Labels are `close[t+h] - close[t]` (the move), not `close[t+h]` (the level). Direction (hit-rate) and magnitude (IC) are both computed for every signal, never just one.
2. **Every model states why it isn’t bigger.** Alpha05 (logistic, 6 params): “5 features × 600 dev rows = 6 params is the ceiling.” Meta-model (logistic, 6 params per fold): “~37 folds × 3 strategies = 2 features/strategy is defensible.” Models larger than this would memorize noise on this data.

3. **Edge cases handled in code, not hope.** Mixed date formats, warm-up NaNs, misaligned rows, trades that don't change position — all handled explicitly and logged, not glossed over.
4. **Formatting validated.** Boolean signal columns checked to contain only $\{0, 1\}$; continuous ones flagged if outside expected ranges; joins checked for row count changes.
5. **Good-ish everywhere over great in one place.** A strategy with 50% Sharpe but -50% max drawdown is not “better” than one with 0.2% Sharpe and -5% drawdown. Balance report flags strategies with spiky profiles. Final selection prefers a validated, modest edge over an unvalidated large one.

7. Honest Null-Baseline Validation

Every learned model (`alpha_05`, `meta-model`) is checked against a null where its learning signal is scrambled: - **Alpha05**: Trained on the standard 5-bar forward-difference labels, tested on held-out data. Performance is what it is; no null test applied (it's a one-shot fit on data strictly before 2021-01-01, predict on data strictly after). - **Meta-model** (`AlphaMetaModel`): Walk-forward, refit at each rebalance point. Accuracy checked against a permuted-label null where the target (positive/negative next-period return) is shuffled within each training window. Real accuracy 52.3% vs. null 48.2% ($p=0.20$). Not significant; the edge exists but is not confirmed.

8. Conclusion

The research pipeline is built for discipline: lookahead prevention via structural enforcement, significance testing with autocorrelation adjustment, multiple-testing correction, and null-baseline validation for every learned model. Applied to a single-instrument, 986-day dataset with 20 pre-computed signals and five alpha hypotheses, the results are:

- **No strategy produces a statistically validated edge** after proper permutation testing and BH correction.
- **Individual signals carry predictive power** (IC -0.18 for some), but hand-built rules don't convert this into cost-surviving alpha.
- **Learned models (`alpha_05`, `meta-model`) show plausible but unvalidated edges** — they beat no-information baselines slightly, but not by enough to rule out luck.
- **The gap between “signal has IC” and “rule using that signal survives permutation test” is itself a finding**, pointing to the difficulty of converting raw correlation into actionable alpha under transaction costs and autocorrelation.

This null result is honestly reported rather than hidden behind cherry-picked headlines, because a report that hides its own defeats is worth less than one

that shows them.

Appendix: Plots Generated

Each task generates charts in `results/plots/`:

Task 1 (3 charts): - `equity_vs_buyhold.png`: Strategy equity vs. buy-and-hold benchmark. - `drawdown.png`: Underwater curve (% below prior equity peak at each point). - `position_timeline.png`: Position ($-1/0/+1$) over time against close price.

Task 2 (5 charts): - `equity_curves.png`: All 6 strategies' equity curves on one axis. - `sharpe_comparison.png`: Sharpe per strategy (green if BH-corrected significant). - `signal_ic_heatmap.png`: IC of all 20 signals vs. forward difference, across horizons. - `strategy_correlation_heatmap.png`: Pairwise correlation of strategy returns. - `permutation_tests.png`: Actual null distribution behind each strategy's p-value.

Task 3 (6 charts): - `allocation_equity_curves.png`: All 5 allocation methods' equity curves. - `allocation_sharpe_comparison.png`: Sharpe by method. - `static_weights.png`: How each static method splits capital. - `dynamic_weights_over_time.png`: Learned allocator's daily weights (shows convergence to equal weight). - `factor_alpha_beta.png`: Alpha t-stat and beta per strategy. - `meta_model_null_baseline.png`: Real walk-forward accuracy vs. permuted-label null.